Module 13

Financial Risk Management

November 24, December 1 & 3, 2025

Market Risk

How should one measure the risk of potential loss due to adverse price movements of assets in one's trading book?

- Variance of asset returns: while it is justifiable if the return distribution is normal (or at least symmetric), it is inadequate if otherwise (e.g., a portfolio of bonds or derivatives).
- Greeks of derivative instruments: perturbative measures, cannot capture risk due to sudden large price movements.

Value at Risk (VaR)

How much a portfolio Π can lose over a given time horizon?

VaR: statistical estimation of such a lose (with a given confidence level α)

$$P(\delta\Pi \le -VaR) = 1-\alpha$$

VaR for a single asset:

To illustrate the idea and also make it analytically tractable, assume the return on an asset S is normal with annualized return rate μ and annualized return standard deviation σ. Over a period δt ,

$$R_{t,t+\delta t} = \frac{\delta S_t}{S_t} = \frac{S_{t+\delta t} - S_t}{S_t} \sim N(\mu\delta t, \sigma^2\delta t)$$

$$P\,\&\,L = S_{t+\delta t} - S_t = S_t R_{t,t+\delta t}$$

$$P(P\,\&\,L \le -VaR) = P(S_t R \le -VaR) = P(R \le -\frac{VaR}{S_t}) = P\left(\frac{R-\mu\delta t}{\sigma\sqrt{\delta t}} \le \frac{-\frac{VaR}{S_t} - \mu\delta t}{\sigma\sqrt{\delta t}}\right)$$

$\frac{R-\mu\delta t}{\sigma\sqrt{\delta t}}$ follows standard normal distribution $N(0,1)$

For a short time horizon δt, ignoring the drift term (as δt is small), for a given confidence level α,

$$P(\delta S_t \le -VaR) = 1-\alpha \rightarrow N^{-1}(1-\alpha) = \frac{-\frac{VaR}{S_t} - \mu\delta t}{\sigma\sqrt{\delta t}} \approx \frac{-\frac{VaR}{S_t}}{\sigma\sqrt{\delta t}},$$

$$\rightarrow VaR = -\sigma S_t \sqrt{\delta t} N^{-1}(1-\alpha)$$

$N^{-1}(.)$ = inverse standard normal distribution function

$\alpha = 99\%$: $N^{-1}(1-\alpha)$ = -2.326342

$\alpha = 97.5\%$: $N^{-1}(1-\alpha)$ = -1.96

$\alpha = 95\%$: $N^{-1}(1-\alpha)$ = -1.644853

Example:

On November 21, 2025, TSLA, BA closed at $391.09 and $179.70 respectively. Their volatility estimated using long (2 year) and short dated (6 month) historical data are 62%, 54% (TSLA) and 35%, 28% (BA) (source: Yahoo Finance).



For 1-day VaR we use annualization factor 250 so that $\sqrt{\delta t} = \sqrt{\frac{1}{250}}$. We compute 1-day VaR:

TSLA	$ 391.09		
vol\alpha	99%	97.50%	95%
62%	35.9002	30.2461	25.3834
54%	30.7962	25.9460	21.7746

BA	$ 179.70		
vol\alpha	99%	97.50%	95%
35%	9.3305	7.8610	6.5972
28%	7.4136	6.2460	5.2418

More general case:

$$VaR = -S_t\left(\mu\delta t + \sigma\sqrt{\delta t}N^{-1}(1-\alpha)\right)$$

Notice that the real world, instead of risk neutral, drift is used above.

<u>VaR for a portfolio of assets</u>:

Consider portfolio

$$\Pi = \sum_{i=1}^{n} w_i S_i \;\rightarrow\; R_\Pi = \frac{\delta\Pi}{\Pi} = \sum_{i=1}^{n} W_i R_i\,,\;\; W_i = \frac{w_i S_i}{\sum_{i=1}^{n} w_i S_i}\;,\;\; \sum_{i=1}^{n} W_i = 1\,. \qquad \text{(see Module 11)}$$

R_i normal → R_Π normal → for a short time horizon Δt:

$$VaR = -\sigma_\Pi \Pi\sqrt{\delta t}N^{-1}(1-\alpha)$$

$$= -\Pi\sqrt{\delta t}N^{-1}(1-\alpha)\sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} W_i W_j \rho_{ij}\sigma_i\sigma_j}$$

$$= -\sqrt{\delta t}N^{-1}(1-\alpha)\sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} w_i S_i w_j S_j \rho_{ij}\sigma_i\sigma_j} \qquad \text{(see Module 11)}$$

Example:

Consider a portfolio consisting 0.5 share of TSLA and 1 share of BA. At current prices the two stocks are weighted at similar level and its current value = 375.25 is comparable to a single share of TSLA. Their correlation estimated from the long and short dated historical data are 0.35 and 0.22 respectively. Using the above data for TSLA and BA we compute 1-day VaR:

corr\alpha	99%	97.50%	95%
0.35	22.9668	19.3497	16.2388
0.22	18.4919	15.5795	13.0748

VaR for a portfolio of derivatives:

Let Π be a portfolio of derivatives with underlying assets $\{S_i\}$, i = 1, 2, …, n. Let

$\Delta_i = \frac{\partial \Pi}{\partial S_i}$, the delta of the entire portfolio w.r.t. S_i

→ $\delta\Pi \approx \sum_{i=1}^{n} \Delta_i \delta S_i = \sum_{i=1}^{n} \Delta_i S_i R_i$

→ $VaR \approx -\sqrt{\delta t} N^{-1}(1-\alpha) \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} \Delta_i S_i \Delta_j S_j \rho_{ij} \sigma_i \sigma_j}$

"≈" because it ignored both higher order terms in δS_i (non-linearity) and changes in other parameters such as interest rate, volatility.

→ to improve VaR calculation one may add additional greeks and higher order terms

For derivative products we have realized and option implied volatility:

- Option valuation → implied volatility
- Asset price movement → forecasted realized volatility, may be different from the option implied volatility

Monte Carlo (MC) VaR:

Widely adopted VaR methodology due to its flexibility in treating various financial assets and products:

- Identify the variables affecting the portfolio, such as
  - asset prices, asset volatilities

- interest rates and shapes of the yield curves
- FX rates
- …

♦ Select pricing models and determine model parameters. For instance apply
- a multivariate lognormal model for asset prices
- a mean-reverting stochastic process for interest rate
- econometric methods to determine their drift rates, covariance matrix
- BSM model for vanilla options
- …

♦ Simulate the market movements to the desired time horizon

♦ Value the portfolio using the selected models at the generated future market scenario

♦ Compute the portfolio P&L by subtracting its current value

♦ Repeat the above process for many random paths to obtain an estimated distribution of the P&L

♦ VaR can be read from this P&L distribution as the $1-\alpha$ quantile of the simulated data

Example:

Simulate 1000 scenarios, order the resulting P&L. For α = 99%, take the last 10th value. For α = 95%, take the last 50th value.

Exercise:

Assume lognormal asset prices, constant interest rate and implied volatility, simulate the VaR of a basket of 2 vanilla options on 2 different assets using BSM formula as the valuation model.

MC VaR is

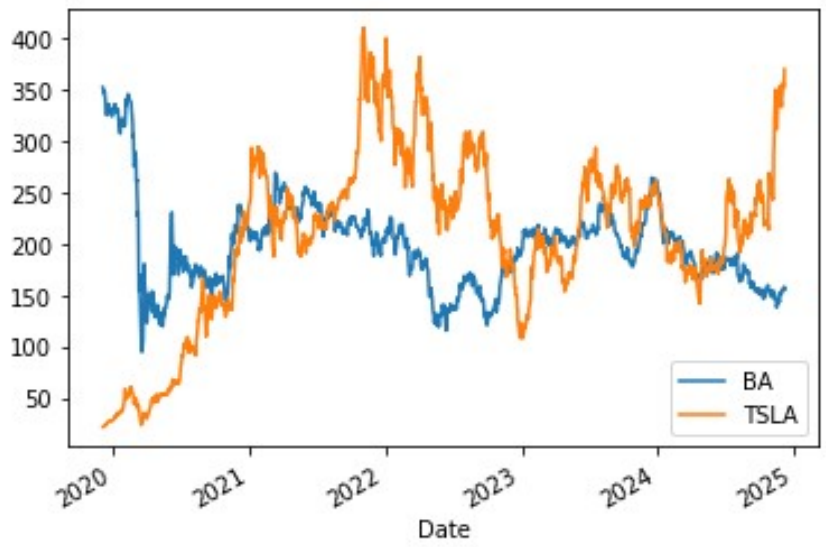
- forward looking – using forecasted parameters
- model dependent – explicit assumptions on the stochastic processes of market variables, P&L on positions of derivatives are estimated with valuation models
- computationally expensive for large portfolios with complex structured products

Historical VaR:

- use historical asset prices instead → assets under consideration need have adequate historical data
- backward looking, which period of the past one should look at?
- less model dependency – no assumptions on market dynamics, no parameters to estimate
- not completely model independent – illiquid derivatives may be "marked to model" historically

Example:

Consider TSLA and BA daily return for the past five years (computed from close prices during November 23, 2020 – November 21, 2025). There are 1255 daily return data points (source: Yahoo Finance).



The historical 1-day VaR computed from these data:

	99%	97.50%	95%
TSLA	26.1800	21.5100	16.3900
BA	12.2500	9.9800	6.9300

For the above portfolio consisting of 0.5 shares TSLA and 1 share of BA, the 1-day VaR is

	99%	97.50%	95%
VaR	19.8950	16.1983	12.5750

Merits for using VaR as a risk measure:

- Consistent risk measure across different risk factors
- Holistic risk measure that takes into account all (identified) risk factors
- Probabilistic risk measure (instead of scenario analysis based on "what if")
- Allow aggregation of risks across all trading books

Criticism for using VaR as a risk measure:

- Failure in capturing tail risk
- Dependence on historical data
- Dependence on pricing models
- …

"Expected shortfall" is a risk measure meant to address the issue of tail risk

Expected Shortfall (ES):

Also called conditional VaR (CVaR), which address the question:

when market fall into the extreme situation indicated by VaR, how much one can lose?

$$ES_\alpha = \frac{1}{1-\alpha}\int_0^{1-\alpha} VaR_\gamma d\gamma\,,\ \ VaR_\gamma = \text{VaR at confidence level } \gamma$$

Example:

Simulate 1000 scenarios, order the resulting P&L. For α = 99%, take average of the last 10 values. For α = 95%, take average of the last 50 values.

Example:

Using the historical data above for TSLA and BA we compute 1-day ES:

	99%	97.50%	95%
TSLA	34.4126	27.9397	23.2292
BA	14.5208	12.4165	10.3168

For the above portfolio consisting of 0.5 shares TSLA and 1 share of BA, the 1-day VaR is

	99%	97.50%	95%
ES	25.0669	20.8420	17.5767

Furthermore ES is also a "coherent risk measure" (while VaR is not):

For portfolio Π_A, Π_B,

$$ES_\alpha(\Pi_A + \Pi_B) \leq ES_\alpha(\Pi_A) + ES_\alpha(\Pi_B)$$

→ ES became more widely adopted after the 2008-2009 financial crisis.

VaR/ES are computed at trader/desk/division/region/firm level to serve as

- benchmark for business – allow comparison of risk in different lines of business
- potential loss estimate, limit on trading books
- reference for setting up capital cushion

Credit Risk

Credit rating:

Major US rating agencies: Moody's, S&P and Fitch:

Moody's	S&P/Fitch	
Aaa	AAA	Investment Grade
Aa	AA	
A	A	
Baa	BBB	
Ba	BB	High Yield
B	B	
Caa	CCC	
Ca	CC	
C	C	

Finer grades between Aa/AA and Caa/CCC: Aa1, Aa2, Aa3/AA+, AA, AA-, etc.

Historical default probability:

Average cumulative default rates (%), 1970–2015 (Source: Moody's):

Term (yrs):	1	2	3	4	5	7	10	15	20
Aaa	0	0.011	0.011	0.031	0.087	0.198	0.396	0.725	0.849
Aa	0.022	0.061	0.112	0.196	0.305	0.54	0.807	1.394	2.266
A	0.056	0.17	0.357	0.555	0.794	1.345	2.313	4.05	6.087
Baa	0.185	0.48	0.831	1.252	1.668	2.525	4.033	7.273	10.734
Ba	0.959	2.587	4.501	6.538	8.442	11.788	16.455	23.93	30.164
B	3.632	8.529	13.515	17.999	22.071	29.028	36.298	43.368	48.071
Caa–C	10.671	18.857	25.639	31.075	35.638	41.812	47.843	50.601	51.319

Average cumulative default rates (%), 1981–2024 (Source: S&P Global):

Term (yrs):	1	2	3	4	5	7	10	12	15
AAA	0	0.03	0.13	0.23	0.34	0.49	0.67	0.73	0.86
AA	0.02	0.05	0.11	0.19	0.28	0.45	0.65	0.76	0.9
A	0.05	0.11	0.19	0.29	0.39	0.65	1.03	1.25	1.56
BBB	0.14	0.38	0.67	1.01	1.36	2	2.86	3.35	4.01
BB	0.56	1.76	3.12	4.48	5.75	7.94	10.44	11.65	13.05
B	2.93	6.93	10.46	13.31	15.6	18.9	22.02	23.43	25.11
CCC/C	26.12	35.92	41.32	44.35	46.53	48.61	50.43	51.32	52.3

Default recovery:

Recover assets from default via

- Direct negotiation (e.g., debt modification, reorganization)
- Bankruptcy court

Amount can be recovered depends on the state of the debtor and the seniority of the debt

Terminology:

- Recovery rate (R): % of asset notional that can be recovered
- Loss given default (LGD)

A model for default:

Let the probability of survival (without default) up to time t is $P_s(0,t)$. Assume that default is a random arrival process with rate λ so that the probability of default between $(t, t+\delta t)$ is $\lambda \delta t$, then the probability of surviving to $t+\delta t$ is

$$P_s(0,t+\delta t) = P_s(0,t)(1-\lambda \delta t)$$

→ $P_s(0,t+\delta t) - P_s(0,t) = -\lambda P_s(0,t)\delta t$

→ $\frac{dP_s(0,t)}{P_s(0,t)} = -\lambda dt$

→ $P_s(0,t) = e^{-\lambda t}$ ($P_s(0,0) = 1$)

λ : "hazard rate" → can be thought as default probability per unit time (default rate).

If λ is time dependent,

$$P_s(0,t) = e^{-\int_0^t \lambda_u du}$$

A simple one-period model for bonds:

Consider a zero coupon bond B with maturity T. Let R be the % can be recovered from the bond notional N, r be the risk free rate (continuous compounding). In a single period setting:

$$B(0,T) = NP_s(0,T)e^{-rT} + NR(1-P_s(0,T))e^{-rT}$$

$$= N\left(e^{-\lambda T} + R(1-e^{-\lambda T})\right)e^{-rT}$$

Example:

A zero coupon bond with notional N = 1000 maturing in 5 years. Assume the risk free rate r = 5%, default (hazard) rate λ = 8%, recovery rate R = 40%

→ DF = 0.7788, Survival Probability = 0.6703

→ bond price = 624.75

One may also deduce default rate λ from the bond market price and recovery rate R.

If B has credit spread s so that $B(0,T) = Ne^{-(r+s)T}$

→ $e^{-sT} = e^{-\lambda T} + R(1-e^{-\lambda T})$

In the above example, s = 4.41%.

When T is small:

$$\lambda \approx \frac{s}{1-R}$$

More general case:

Consider the case where default can happen at anytime between $(0,T)$. Assume R is constant, independent from the timing of default, and the recovery is instantaneous. Assume also that λ is independent from r.

$$B(0,T) = NP_s(0,T)e^{-\int_0^T r_u du} + NR\int_0^T \lambda_t P_s(0,t)e^{-\int_0^t r_u du} dt$$

$$= Ne^{-\int_0^T (\lambda_u + r_u) du} + NR\int_0^T \lambda_t e^{-\int_0^t (\lambda_u + r_u) du} dt$$

1st term: probability of survival till maturity × PV of the notional

2nd term: sum of (probability of surviving till t but default during $(t, t+dt)$ ×

PV of recovery value)

Caveats:

- Recovery is not instantaneous in practice: bankruptcy proceeding could be a lengthy process

- Amount can be recovered may be negatively correlated with likelihood of default
- λ and r may be (correlated) stochastic variables

How to determine λ ?

Assume R can be estimated through fundamental analysis, then $\{B, R\} \rightarrow \lambda$

→ "reduced form model"

Once λ_t is determined from a set of liquid instruments (bonds, CDS, …) one can use it to price other credit sensitive instruments (such as illiquid debts, credit derivatives, etc).

Firm Value (FV) model (Merton, ...)

FV = asset + liability

At maturity of liability

- asset > liability → shareholders pay off or refinance debts
  → firm's business continues beyond the maturity date
- asset < liability → shareholders walk off the debt obligation
  → bankruptcy liquidation, debt holders collect remains of the firm

→ Equity = call option on FV

Firm value V_F may be modeled as a stochastic variable, e.g., a lognormal process

$$\frac{dV_F}{V_F} = \mu_F dt + \sigma_F dW_t$$

→ Valuing call on FV in the same way as pricing call on equity

equity price, volatility → FV, FV volatility

→ value of debts, including convertible bonds

→ Elegant, unified treatment across capital structure (senior, junior debts, convertible bonds, preferred stocks, common equity) of the firm

→ "structural model'

Example:

Assume the firm is capitalized with common equity S and debt B maturing at time T

$V_F = V_S + V_B$

Here V_S and V_B represent the aggregated amount of equity and bond respectively. Let N_B be the aggregated notional of bond.

Payoff at time T:

$$\begin{cases} V_B = V_F & V_F < N_B \\ V_B = N_B & N_B \le V_F \end{cases}, \quad \begin{cases} V_S = 0 & V_F < N_B \\ V_S = V_F - N_B & N_B \le V_F \end{cases}$$

- share holders ➔ long a call on FV with strike N_B
- debt holders ➔ long FV + write the call

  ➔ long debts + short a put on FV

Assume S pays no dividend and B is a zero coupon bond, apply the BSM to V_S:

$V_S = V_F N(d_1) - N_B e^{-r(T-t)} N(d_2)$,

$\sigma_S V_S = \frac{\partial V_S}{\partial V_F} \sigma_F V_F = N(d_1)\sigma_F V_F$, (Ito's lemma applied to $V_S(V_F, t)$)

where r is risk free rate, σ_S is the equity volatility,

$$d_1 = \frac{\ln\left(\frac{V_F}{N_B}\right) + \left(r + \frac{\sigma_F^2}{2}\right)(T-t)}{\sigma_F \sqrt{T-t}}, \quad d_2 = d_1 - \sigma_F \sqrt{T-t}$$

➔ $\{V_S, \sigma_S\} \to \{V_F, \sigma_F\} \to V_B \qquad (\to P_s(0,T) \to \lambda)$

(See Hull 24.6 (p 548-549) for a numerical example)

It becomes more complicated with increasing complexity of the capital structure of the firm, e.g., existing senior debts or other convertibles, debts with different maturities

A simple refinement on default timing:

Assume the firm may default prior to T when $V_F(t) < L$, where L can be either greater or less than N_B

➔ equity = KO option on FV where L may be deduced from equity option volatility skew

(option on equity ➔ compound option on FV)

Further refinements and applications of structural models:

- Moody's KMV: "distance to default"
- Optimal capital structure of the firm, optimal corporate finance policy
- "Capital structure arbitrage"

Types of credit risk:

1) credit risk inherited in security valuations such as corporate bonds

   → can be treated as parts of "market risk"

2) lending risk: amount is known, bilateral agreement but risk is one sided

3) counterparty risk: bilateral risk, amount is uncertain

   Example: consider a long forward contract f with entity A on a non-dividend paying stock S with forward price K and maturity T:

   $f_t = S_t - Ke^{-r(T-t)}$

   $S_t > Ke^{-r(T-t)}$ → counterparty risk

   $S_t < Ke^{-r(T-t)}$ → no counterparty risk (but A has counterparty risk with you)

   → Amount at risk varies with S_t

Credit exposure:

Credit exposure (CE) is the amount one stands to lose in the event of counterparty default. It depends on both time horizon and future market scenarios.

Credit exposure = max(position value, 0)

- Potential future exposure (PFE, for a given confidence level)

  Similar to VaR, PFE of a portfolio of bilateral trades Π at a given confidence level α may be expressed as

$$P\left(CE_{\Pi}(t) \geq PFE\right) = 1-\alpha$$

Similar to VaR, PFE is usually computed via Monte Carlo simulations.

PFE is evaluated at trade, counterparty, portfolio and firm level to assess both potential credit exposure to specific entities and firm wide counterparty risk:

- Limit on trades, at counterparty and portfolio level
- reference for setting up capital reserve for credit loss

Exercise:

Compute PFE with confidence level α at $t < T$ for a long forward contract maturing at T seen from time $t_0 < t$, assuming asset S follows a lognormal process.

- Expected exposure (EE)

Example:

EE at time $t < T$ for a long forward contract seen from time $t_0 < t$: assume S follows a lognormal process so that

$$S_t = S_{t_0} e^{\left(\mu - \frac{\sigma^2}{2}\right)(t-t_0)+\sigma W_t}$$

$$EE = E_{t_0}\left[\max(S_t - Ke^{-r(T-t)}, 0)\right]$$

$$= E_{t_0}\left[\max(S_t - Ke^{-r(T-t)}, 0)\right] e^{-\mu(t-t_0)} e^{\mu(t-t_0)}$$

$$= c(S_{t_0}, t_0, Ke^{-r(T-t)}, t) e^{\mu(t-t_0)} \qquad \text{(c is priced under drift rate } \mu\text{)}$$

$$= S_{t_0} N(d_1) - Ke^{-r(T-t)} e^{-\mu(t-t_0)} N(d_2)$$

$$d_1 = \frac{\ln\left(\frac{S_{t_0}}{Ke^{-r(T-t)}}\right) + \left(\mu + \frac{\sigma^2}{2}\right)(t-t_0)}{\sigma\sqrt{t-t_0}}, d_2 = d_1 - \sigma\sqrt{t-t_0}$$

Note: here the expectation $E[.]$ is taken w.r.t. the real world probability measure instead of the risk neutral probability as in option pricing.

Credit Valuation Adjustment (CVA):

Also called counterparty value adjustment, it is the cost of hedging default risk of the counterparty of the transaction.

Let R be the recovery rate, EE be the expected exposure of the derivative transaction (maturing at T) in question, λ_t be the counterparty default intensity, r the risk free interest rate, trade notional $N = 1$,

$$CVA(t_0) = \int_{t_0}^{T} (1-R) EE(t_0,t) \lambda_t P_s(t_0,t) e^{-\int_{t_0}^{t} r_u du} dt = (1-R) \int_{t_0}^{T} EE(t_0,t) \lambda_t e^{-\int_{t_0}^{t} (\lambda_u + r_u) du} dt$$

In the above it is implicitly assumed that R is constant and counterparty default is uncorrelated with the derivative valuation and interest rate level.

Valuation Adjustments (xVA) on Financial Engineering Transactions:

- CVA
- Debit value adjustment (DVA) - CVA from the counterparty's perspective
- Funding value adjustment (FVA) - adjustment due to funding cost/benefit
- Collateral value adjustment (ColVA) - adjustment due to collateral posting
- Margin value adjustment (MVA) - funding costs of the initial margin
- Capital value adjustment (KVA) - cost of holding regulatory capital

→ Price of an OTC derivative = base price + xVA

Mitigating counterparty risk:

Counterparty risk is important when one has a large OTC derivative portfolio. To mitigate counterparty risk one may resort to

- Netting
- Collateral
- Contractual clauses
- Hedging

- CVA desk:
    - Price, hedge or own the counterparty risk
    - Expanded to xVA: xVA components may overlapping, need holistic and consistent approaches
- Central counterparty (CCP)

Scenario Analysis and Stress Tests

Comprehensive Capital Analysis and Review (CCAR):

Conducted annually by large US banks and financial institutions (submitted to and reviewed by Fed):

Forward looking, model based analysis and test under

- baseline scenario
- adverse scenario
- severely adverse scenario

Based on

- market risk models
- credit risk models
- operational risk models
- aggregation models
- …

Dodd–Frank Act Stress Testing (DFAST):

Similar tests required for smaller banks